EXNO: 3 Designing a Quarter wave transformer by varying

following parameters

a) frequency

b) input impedance

c) characteristic impedance

Test Case 1:

Design a quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

AIM: To design a quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms and to analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz }

SOFTWAREREQUIRED:

SCILAB.

PROGRAM:

```
clear;clc;

Zr=100;s=9;d=0.1;

Zs=50;

Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;

Zof=276*log10(s/r);

Zoq=sqrt(Zr*Zof);

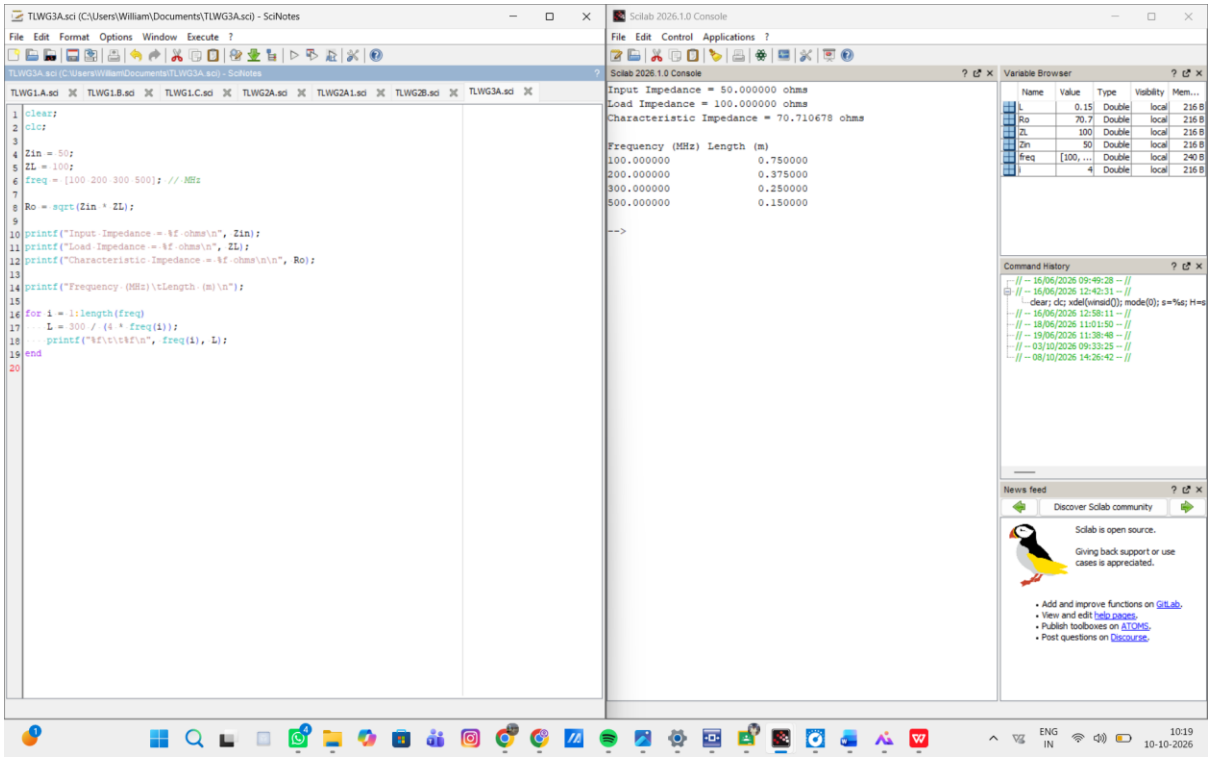
L=300/(f*(10^-6));

printf("length of lambda/4 transformer = %f meters\n",L);

printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```

OUTPUT:

ZR=100 ohms Zin =50 ohms



TABULATION:

Frequency (MHz)	Length of λ/4 transformer (m)
100	0.750
200	0.375
300	0.250
500	0.150

RESULT:

A quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms is designed and the transformer's performance is analyzed for the varying operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz }.

Test Case 2:

For a fixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance (Zin) varies through {25 ohms, 50 ohms, 75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance (Ztransformer= Ro= sqrt(Zin·ZL) changes with Zin.

AIM: To design a quarter-wave transformer a ftixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms and evaluate its performance as the input impedance (Zin) varied. To validate that the transformer's characteristic impedance (Z transformer= Ro= sqrt(Zin·ZL) changes with Zin.

SOFTWAREREQUIRED:

SCILAB

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Typetheprogram

● Execute the program

● End the program

PROGRAM:

```
clear;clc;

Zr=100;s=9;d=0.1;

Zs=50;

Ro=sqrt(Zs*Zr);f=100*(10^6);r=d/2;

Zof=276*log10(s/r);

Zoq=sqrt(Zr*Zof);

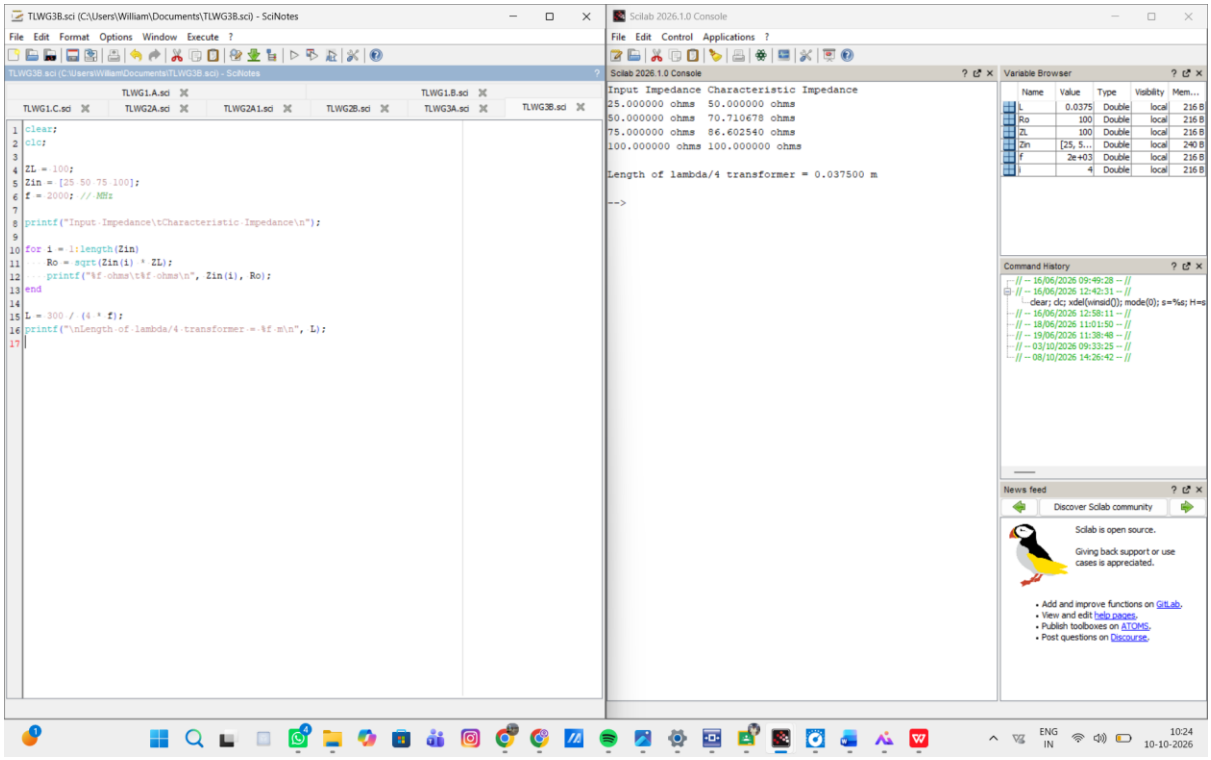
L=300/(f*(10^-6));

printf("length of lambda/4 transformer = %f lambda\n",L);

printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```

OUTPUT:

ZR=100 ohms Variable :Zin



TABULATION:

Input impedance $Z_{in}(\Omega)$	Load impedance $Z_L(\Omega)$	Characteristic impedance $R_o(\Omega)$
25	100	50.000
50	100	70.711
75	100	86.603
100	100	100.000

For $Z_L = 100\ \Omega$:

Characteristic impedance $R_o(\Omega)$	Load impedance $Z_L(\Omega)$	Input impedance $Z_{in}(\Omega)$
50	100	25
60	100	36
70	100	49
80	100	64
100	100	100

RESULT:

A quarter-wave transformer is designed at a fixed operating frequency of 2 GHz and load impedance (ZL) of 100 ohms and its performance is evaluated for various input impedance values. The variation of transformer's characteristic impedance with Zin is also validated..